AI Model Used: **Gemini, Claude**

What AI Was Used For:

AI was used for:

- Project planning, breaking down requirements, finding and selecting node.js libraries
- Getting started on the code for the RSA key generation in keyManager.js and the Express server in server.js
- Refactoring and updating my code based on the gradebot test client results
- Generating most of the test suite to meet the coverage requirements
- Structuring, updating, and revising the README
- Revising git commit messages

Prompt 1:

A month ago i completed project 1 for my foundations of cybersecurtiy class which was a simple jwks server. I attached pdfs of the assignment instructions and a notes google doc i made to help me with the project. I attached the instructions for project 2 and a template for my notes doc for project 2; we're extending the project by using a SQLite single-file database. I'm not familiar with sqlite i've never used it. Can you help me fill out the notes doc before i get started on the code for this project? I only need to add one file, database.js, to the project right? What exactly will this extension of the project do? And what will the test client do to test the project?

Prompt 2:

Read all the files in C:\Users\mxz\Desktop\Projects\3550-project2-documents. I'm currently working on project 2 for my foundations of cybersecurity class. I just created database.js and added some code to initalize the database and create a table matching the schema in the assignment instructions. First make a brief commit summary message for this change. Then, can you make a list of all the specific code changes I need to make, functions i need to create, etc. in database.js, keyManager.js, and server.js for project 2?

Prompt 3:

Ok i was working on database.js. First off, idk what u meant by let an environmental variable override the database path. I tried to implement the rest of the changes you suggested. Can you review the changes i made, and how do i verify these changes work/are good before i commit them and start working on keyManager.js to finish the "2. Save private keys to the DB" requirement for the assignment.

Prompt 4:

i wrote some code in keyManager.js. u told me to compute exp but i'm pretty sure expiry already has the value i need so i just used expiry wwhen calling insertkey. The privatekey is already generated in pem format, so it was safe to save it in the database like that. however, i'm working on rowToJWK, but idk what the purpose of this function is, where it's going to be called, and how to implement it. can you help me

Prompt 5:

ok i'm on server.js. I updated the get endpoint, and now i'm on the post/auth endpoint. On the auth endpoint, i selected the key using database.js's getKey(), but now i'm unsure howw to sign the token since the key is now targetKeyRow, a row from the table instead of a key object i think. Can you review my changes and help me with this

Prompt 6:

PS C:\Users\mxz\Desktop\Projects\jwks-server> curl.exe -X POST http://localhost:8080/auth

eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCIsImtpZCI6IjMifQ.eyJ1c2VyIjoibW9ja191c2VyIiwiZXhwIjoxNzkyOTU2MzYzLCJpYXQiOjE3OTI5NTMxMzR9.jhQu9qTjO6VHrAdLs8mWKH_EMyQZreNU-2Y3H2uWtvyvRGY3NTK5n9YnNIXAsiTPRgwg0AzDRUa-Mdig8_q9M1_dSWCYbMSMIIk6uL-hFe_CGXPm_YJyMDZWC8v0t2KqWU9td7T_54BlwWdm4w8XA947khcfwC7nm6n7tJjXDQnp225Culhc380sIs33wbzgyGYB4ZwuqVdXVOvLiquDsSqx5uzBH426Yaohvu5aZHkBkQniEZdxk4wIe1gqQREVQTf89pHIvQUY16mrng63u82lwcccOvY-xWHUQ6TbbM-fAadPHZExK-XANp0cKDga7AHMzwG3pOc-IuqCTiQQWw

PS C:\Users\mxz\Desktop\Projects\jwks-server> curl.exe -X POST "http://localhost:8080/auth?expired=true"

eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCIsImtpZCI6IjEifQ.eyJ1c2VyIjoibW9ja191c2VyIiwiZXhwIjoxNzkyOTAyMDMwLCJpYXQiOjE3OTI5NTMxNTZ9.hLgMCK-3kOBurBhW_Rb09o9CgZXeQ7aK_xvNnREdHEyp2KXSwIfztvzDze9vSYP_VN2Drd-zunaowcIFIeKIkT94CF8zBOH0XG0xBxlIWHItcn0qFX1vc53bVRWRqSh4EYgsQ0gmTcVQTNQniJH2Rmi5hRhdYDkGdl2SCrxDHeKT-QcMhXEho-S8BykPOwwXOmZTDqye6-T9kB_CYMNf5F6GfaurRgnpF1yfjuBc5JROAPPyERgnGZ2d7Gt1vS5aaZHY6KDP2G3-wKmLZySuPik41q6c5wMaDhdiKMNTIGl5Ylq9_6o4L5zL_Yf6Exzi9hqH9rz-l0QspMx7KJzqBA

PS C:\Users\mxz\Desktop\Projects\jwks-server> curl.exe http://localhost:8080/.well-known/jwks.json

{"keys":[{"kty":"RSA","n":"m0IJE43Nfk94OGTA8hOPJbjSjuqjHG1PeMciLYrddOGSHfZLjemUugyRh8dnvT5md-fM1Zwxi7KsvZahcM8hlX_C0zhj9eH26cHDrOttsGFelvDQR9viYvUg_REjzN8IcJr4pCNmZCGqWykCaRb3kUu-ba7vJM7J5vH2IIeFtMerywUjXEcOrYUGsu4_shfKSezRD90OgYKMPVZpQA5duDeOG74-czJUymIgTFPGdogoPWdh6cBy1zu_QySgJyVvlvebQaZvklZybOzd7Dh3GGoMJsX9y2C8nPH_PDq4nPJaLDBJDghWUHS5ZZ8m3dJLI7dJeYr6_3YBUY6yfiY-1Q","e":"AQAB","kid":"3","alg":"RS256","use":"sig"}]}

PS C:\Users\mxz\Desktop\Projects\jwks-server>

can you explain what each of these 3 commands and outputs do/show

Prompt 7:

Can you read "C:\Users\mxz\Desktop\Projects\3550-project2-documents\project2-notes.pdf" again. I just ran the gradebot client and it gave me a 98 which i'm satisfied with but i still need to

update the tests (keyManager.test.js, server.test.js), and make the tests for database.test.js. Can you help me update/make these tests?

Prompt 8:

PS C:\Users\mxz\Desktop\Projects\jwks-server> npm test

> jwks-server@1.0.0 test
> jest --coverage --forceExit

'jest' is not recognized as an internal or external command,
operable program or batch file.
PS C:\Users\mxz\Desktop\Projects\jwks-server> npm ci
npm warn deprecated glob@10.5.0: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm error code 1
npm error path C:\Users\mxz\Desktop\Projects\jwks-server\node_modules\better-sqlite3
npm error command failed
npm error command C:\Windows\system32\cmd.exe /d /s /c node-gyp rebuild
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@12.4.0
npm error gyp info using node@26.8.2 | win32 | x64
npm error gyp info find Python using Python version 3.14.7 found at "C:\Users\mxz\AppData\Local\Python\pythoncore-3.14-64\python.exe"
npm error gyp http GET https://nodejs.org/download/release/v26.8.2/node-v26.8.2-headers.tar.gz
npm error gyp http 200 https://nodejs.org/download/release/v26.8.2/node-v26.8.2-headers.tar.gz
npm error gyp http GET https://nodejs.org/download/release/v26.8.2/SHASUMS256.txt
npm error gyp http GET https://nodejs.org/download/release/v26.8.2/win-x64/node.lib
npm error gyp http 200 https://nodejs.org/download/release/v26.8.2/win-x64/node.lib
npm error gyp http 200 https://nodejs.org/download/release/v26.8.2/SHASUMS256.txt
npm error gyp ERR! find VS
npm error gyp ERR! find VS --msvs_version was not set on the command line
npm error gyp ERR! find VS VCINSTALLDIR not set, not running in VS Command Prompt
npm error gyp ERR! find VS could not use PowerShell to find Visual Studio 2017 or newer, try re-running with '--loglevel silly' for more details.
npm error gyp ERR! find VS
npm error gyp ERR! find VS Failure details: undefined
npm error gyp ERR! find VS could not use PowerShell to find Visual Studio 2017 or newer, try re-running with '--loglevel silly' for more details.
npm error gyp ERR! find VS
npm error gyp ERR! find VS Failure details: undefined
npm error gyp ERR! find VS not looking for VS2017 as it is only supported up to Node.js 21

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  npm error gyp ERR! find VS not looking for VS2017 as it is only supported up to Node.js 21
  npm error gyp ERR! find VS not looking for VS2017 as it is only supported up to Node.js 21
  npm error gyp ERR! find VS not looking for VS2015 as it is only supported up to Node.js 18
  npm error gyp ERR! find VS not looking for VS2013 as it is only supported up to Node.js 8
  npm error gyp ERR! find VS
  npm error gyp ERR! find VS ************************************************************
  npm error gyp ERR! find VS You need to install the latest version of Visual Studio
  npm error gyp ERR! find VS including the "Desktop development with C++" workload.
  npm error gyp ERR! find VS For more information consult the documentation at:
  npm error gyp ERR! find VS https://github.com/nodejs/node-gyp#on-windows
  npm error gyp ERR! find VS ************************************************************
  npm error gyp ERR! find VS
  npm error gyp ERR! configure error
  npm error gyp ERR! stack Error: Could not find any Visual Studio installation to use
  npm error gyp ERR! stack at VisualStudioFinder.fail (C:\Program
Files\nodejs\node_modules\npm\node_modules\node-gyp\lib\find-visualstudio.js:118:11)
  npm error gyp ERR! stack at VisualStudioFinder.findVisualStudio (C:\Program
Files\nodejs\node_modules\npm\node_modules\node-gyp\lib\find-visualstudio.js:74:17)
  npm error gyp ERR! stack at process.processTicksAndRejections
(node:internal/process/task_queues:104:5)
  npm error gyp ERR! stack at async createBuildDir (C:\Program
Files\nodejs\node_modules\npm\node_modules\node-gyp\lib\configure.js:112:18)
  npm error gyp ERR! stack at async run (C:\Program
Files\nodejs\node_modules\npm\node_modules\node-gyp\bin\node-gyp.js:81:18)
  npm error gyp ERR! System Windows_NT 10.0.26200
  npm error gyp ERR! command "C:\\Program Files\\nodejs\\node.exe" "C:\\Program
Files\\nodejs\\node_modules\\npm\\node_modules\\node-gyp\\bin\\node-gyp.js" "rebuild"
  npm error gyp ERR! cwd
C:\Users\mxz\Desktop\Projects\jwks-server\node_modules\better-sqlite3
  npm error gyp ERR! node -v v26.8.2
  npm error gyp ERR! node-gyp -v v12.4.0
  npm error gyp ERR! $npm_package_name better-sqlite3
  npm error gyp ERR! $npm_package_version 13.0.3
  npm error gyp ERR! not ok
  npm error A complete log of this run can be found in:
C:\Users\mxz\AppData\Local\npm-cache\_logs\2026-10-25T19_34_46_679Z-debug-0.log
  PS C:\Users\mxz\Desktop\Projects\jwks-server> npm test

  > jwks-server@1.0.0 test
  > jest --coverage --forceExit

  node:internal/modules/cjs/loader:1573
    throw err;
    ^
```

```
  Error: Cannot find module
'C:\Users\mxz\Desktop\Projects\jwks-server\node_modules\jest\bin\jest.js'
      at Module._resolveFilename (node:internal/modules/cjs/loader:1569:15)
      at wrapResolveFilename (node:internal/modules/cjs/loader:1123:27)
      at defaultResolveImplForCJSLoading (node:internal/modules/cjs/loader:1147:10)
      at resolveForCJSWithHooks (node:internal/modules/cjs/loader:1174:12)
      at Module._load (node:internal/modules/cjs/loader:1346:5)
      at wrapModuleLoad (node:internal/modules/cjs/loader:261:19)
      at Module.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:154:5)
      at node:internal/main/run_main_module:33:47 {
    code: 'MODULE_NOT_FOUND',
    requireStack: []
  }

  Node.js v26.8.2
  PS C:\Users\mxz\Desktop\Projects\jwks-server>
```

Prompt 9:

ok now can you, in a new docs branch, do the jsdoc and comment pass (update all the jsdoc comments and other comments in all the files) and update the readme. for the readme, update the features, tech stack, prerequisites and anything else that needs updating. for the test client section, keep the project 1 command and add the project 2 command: ./gradebot project-2 --port 8080 --code-dir . --database totally_not_my_privateKeys.db --run "node server.js". For the screenshots keep the project 1 screenshots section and add a section for projct 2 screenshots, i haven't taken them yet.